# A Frozen-Prefix Proof-Completion Experiment

## Induction on Proof Lines in *Depths of a Simulation* 45.2

An in-chat AMLD+0-style settlement, source comparison, and mathematical audit

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Technical drafting and in-chat proof generation: DATA MIND / ChatGPT

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**Outcome in one sentence.** An in-chat, AMLD+0-style role representation produced a complete, mathematically sound *candidate* proof of an induction principle from a frozen partial proof. The continuation followed the original argument; the original ending had appeared in this conversation before the attempt. This was **not** a valid blind test and produced **no authenticated AMLD+0 execution or independent machine-verifier acceptance receipt**.

**Abstract**

We report a proof-completion experiment centered on the *Induction Principle on Proofs* from Brian Tenneson's *Depths of a Simulation* 45.2. The researcher selected a theorem and required the entire admissible history to be represented as one conjunction of the theorem's assumptions and the actual proof prefix, with induction-step assumptions properly scoped. The handoff was frozen after the sentence asserting that all earlier proof lines belong to an inference-closed set. A revision moved the cutoff earlier than an initially discussed 75% split: 51 of 100 whitespace-delimited words of the source proof were visible. A subsequent armed prompt asked an in-chat AMLD+0-style settler to finish the proof, permitting a different valid argument. The response completed the proof by ordinary induction, and a later comparison found it mathematically sound and methodologically equivalent to the source. We carefully distinguish mathematical completion, an in-chat role simulation, a first-person future-tense settlement claim, a genuine blind discovery trial, and independently certified execution. Because this conversation previously contained the full source proof, and because no external governing verifier, authenticated solver run, frozen budget, or measured cost ledger was established, the episode documents a *sound reconstruction demonstration and an informative protocol rehearsal*, not independent AMLD+0 discovery or a certified empirical performance result. The note preserves the successful mathematical argument, identifies the audit boundaries, and states the minimum conditions for a future controlled replication.

**Keywords:** automated theorem proving; AMLD+0; proof completion; finite inference systems; proof induction; frozen prefixes; verifier gating; experimental provenance; source exposure; epistemic self-reference.

## 1 Why this experiment was attempted

The starting idea was to create an *artificial basin* without waiting for a proof-search system to become stuck. A completed mathematical argument is deliberately cut at a predetermined location. The settler receives the exact legitimate history up to that location and is asked to finish the proof. This generalizes a previous partial-proof handoff concept and leaves the solver free either to reconstruct the missing reasoning or to discover a different correct proof. The interruption itself is not evidence that the original prover encountered an actual basin.

The researcher first considered a three-quarter proof handoff, then chose a *shorter and earlier* cutoff. The precise proof was selected from the actual integrated 45.2 manuscript, rather than from the separate final-audit PDF. This source distinction is essential: the four-page audit reports on the manuscript's mathematical state but is not itself a numbered, machine-checked derivation of every theorem [1, 2].

The experiment's first-person conjecture was fixed as

$$\boxed{\text{“I will finish the proof.”}} \tag{1}$$

Here “I” was intended to refer to the designated theorem-proving subject, potentially DATA MIND operating through AMLD+0, *not* to the human researcher. The semantics intended an *actual future proof completion*, not a bare assertion of ability or the logical existence of some possible proof. No separately authenticated machine identity was eventually bound to the in-chat demonstration.

## 2 Frozen mathematical task and available background

Let $F = (x, y, z)$ be a formal system with a set $y$ of well-formed formulas and finitary inference rules $z$. For $\Gamma \subseteq y$, write

$$\mathrm{Con}_F(\Gamma) = \{\psi \in y : \Gamma \vdash_F \psi\}$$

for the set of consequences with finite proofs from $\Gamma$. A set $W \subseteq y$ is *inference-closed* when every direct consequence of premises drawn from $W$ also belongs to $W$.

**Theorem 1** (Induction principle on proofs; source target)**.** *If $\Gamma \subseteq y$, $W \subseteq y$, $\Gamma \subseteq W$, and $W$ is inference-closed, then*

$$\boxed{\mathrm{Con}_F(\Gamma) \subseteq W.} \tag{2}$$

This is induction on *proof lines*, not structural induction on a grammar of WFF formation. No grammatical least-generation assumption is needed for Theorem 1.

### 2.1 What the “giant conjunction” meant

The researcher requested that the entire legitimate proof history, plus *all hypotheses in the theorem statement*, be admitted as one conjunction. For arbitrary $\psi \in \mathrm{Con}_F(\Gamma)$ and a finite proof $\pi = \langle \pi(1), \ldots, \pi(n) \rangle$ from $\Gamma$ with $\pi(n) = \psi$, the recorded state had the schematic form

$$\begin{aligned} H_{\text{known}} := \ & \mathrm{FormalSystem}(F) \wedge (\Gamma \subseteq y) \wedge (W \subseteq y) \\ & \wedge (\Gamma \subseteq W) \wedge \mathrm{InferenceClosed}_F(W) \\ & \wedge (\psi \in \mathrm{Con}_F(\Gamma)) \wedge \mathrm{ProofFrom}_F(\pi, \Gamma) \\ & \wedge (\pi = \langle \pi(1), \ldots, \pi(n) \rangle) \wedge (\pi(n) = \psi) \\ & \wedge (\pi(1) \in \Gamma) \wedge (\pi(1) \in W) \\ & \wedge \Big[ 2 \le j \le n \wedge \bigwedge_{i=1}^{j-1} (\pi(i) \in W) \Big]_{\text{local step}}. \end{aligned} \tag{3}$$

The last bracket is a *locally scoped induction hypothesis*, to be discharged in proving the inductive step. It cannot be used as an unrestricted global axiom. The witness $\psi, \pi$ is likewise selected for an arbitrary consequence, not preloaded as a proof of the theorem. In particular, neither $\pi(j) \in W$ for the current unknown line, nor $\psi \in W$, nor the target inclusion (2) was an admitted premise.

Definitions of proof, rule application, and closure were available as part of the source formal-system framework; no conclusions from unrelated manuscripts were silently promoted to new axioms.

# 3 Source selection, cutoff, and freeze

## 3.1 The exact visible prefix

The integrated 45.2 source contains the following proof text up to the frozen boundary [1]:

> Let $\psi \in \mathrm{Con}_F(\Gamma)$. Choose a proof $\pi = \langle \pi(1), \ldots, \pi(n) \rangle$ from $\Gamma$ with $\pi(n) = \psi$. I claim that every line of $\pi$ belongs to $W$.
>
> Proceed by induction on the line number. If $j = 1$, then $\pi(1) \in \Gamma \subseteq W$. Suppose now that $\pi(1), \ldots, \pi(j-1) \in W$.

**Frozen boundary:** immediately after *"Suppose now that* $\pi(1), \ldots, \pi(j-1) \in W$*."* None of the author's subsequent words were included in the canonical frozen handoff.

The source proof is unnumbered mathematical prose. Therefore a statement that "75% of the verified proof steps" was supplied would be inaccurate. The frozen manifest records **51 visible words out of 100**, using whitespace-delimited source proof tokens, i.e., **51% by that counting convention**. Neither this word fraction nor the manuscript audit represents a machine-checked proof-step count.

## 3.2 Freeze identity and provenance

The stable SHA-256 of the actual integrated TeX source was:

```
f44cc417f38a9818dbffcbc415da2d27e5130ca31e5701c0515614ad323954a8
```

The canonical visible prefix and withheld suffix were recorded in a separate freeze manifest. The source-visible prefix digest was:

```
3b10ead33cafadf2a5c71b1e8eb376d4d83917f579d217787624d9baffe9feb0
```

These digests establish that specific files and text segments were frozen; they do *not* prove that an external prover saw only the prefix. A hash is a provenance aid, not an execution trace or verifier receipt.

# 4 What was actually submitted and done

1. The theorem was chosen from the integrated Version 45.2 source and the cutoff moved earlier at the researcher's request.
2. The researcher's additional constraint was incorporated: the theorem's own assumptions and all legitimate prior proof history were made available as one conjunction, with proper local scope for the induction hypothesis.
3. An *unarmed* frozen handoff was prepared, followed by an *armed* prompt. The first armed version required an authenticated executor and governing verifier and consequently produced a NOT_STARTED response in this chat.
4. A subsequent prompt explicitly requested an *in-chat AMLD+0-style* attempt, with roles designated Player-0, P, R, D, I, and C; the researcher pasted that prompt into the *same* conversation.

5. The in-chat response constructed a seven-part mathematical proof, with a role-assessment table. The mathematical proof was evaluated in prose, not by a separately invoked certified solver or independently run machine checker.

6. After the result, the researcher supplied the final mathematical audit and asked for comparison. The assistant compared the candidate argument to the original manuscript and reported that it was sound, substantially the same proof, not genuinely blind, and not machine-certified.

The pasted prompt's mathematical typesetting had some escaped-character and subscript corruption in transport through chat. Its intended formulas were recoverable from the nearby prose, the earlier frozen source, and ordinary mathematical notation. It was *not* parsed or executed as a formal input language; the normalized display here must not be confused with byte-for-byte replay of the pasted text.

## 5 Candidate continuation and mathematical review

The assistant's completed argument used ordinary strong induction on proof-line number. Here it is restated without implying formal-machine acceptance.

*Candidate completed argument (mathematical reconstruction).* Fix an arbitrary $\psi \in \mathrm{Con}_F(\Gamma)$. By definition of consequence, choose a finite proof $\pi = \langle \pi(1), \ldots, \pi(n) \rangle$ from $\Gamma$ with $\pi(n) = \psi$. We show by induction on $j \in \{1, \ldots, n\}$ that $\pi(j) \in W$.

For $j = 1$, because the source proof rules have positive arity, the first line has no earlier rule premises; it is a hypothesis, so $\pi(1) \in \Gamma \subseteq W$. Now fix $2 \leq j \leq n$ and suppose $\pi(i) \in W$ for all $i < j$. By the definition of a formal proof, there are two cases.

If $\pi(j) \in \Gamma$, then $\pi(j) \in W$ by $\Gamma \subseteq W$. Otherwise there are a legal finitary rule $r \in z_k(F)$ and indices $i_1, \ldots, i_k < j$ such that

$$\pi(j) = r\big(\pi(i_1), \ldots, \pi(i_k)\big).$$

Every rule premise belongs to $W$ by the induction hypothesis. Inference-closure of $W$ therefore gives $\pi(j) \in W$. The induction step holds in either case.

Thus every line of $\pi$ belongs to $W$, in particular $\psi = \pi(n) \in W$. Since $\psi \in \mathrm{Con}_F(\Gamma)$ was arbitrary, $\mathrm{Con}_F(\Gamma) \subseteq W$. □

### 5.1 Step-by-step dependency audit

| Candidate step | Justification and original correspondence |
|---|---|
| Arbitrary $\psi$ and proof $\pi$ | Definition of $\mathrm{Con}_F(\Gamma)$; same opening in source. |
| Induction and base case | Positive-arity finite-proof definition; same induction in source. |
| Hypothesis-line case | $\pi(j) \in \Gamma \subseteq W$; same source case. |
| Inference-line case | Premises occur earlier and lie in $W$ by the local induction hypothesis; the candidate makes rule indices explicit. |
| Use of closure | $W$ inference-closed implies the rule's conclusion is in $W$; same original reasoning. |
| Final generalization | $\pi(n) = \psi \in W$, arbitrary $\psi$; identical logical conclusion. |

Table 1: Comparison of the candidate argument with the original source proof.

**Mathematical finding.** The candidate proof is sound under the definitions supplied in the manuscript, including finitary positive-arity rules and the local induction hypothesis. It does not assume the target. It is *not textually identical* to the original; it spells out earlier line indices and the case analysis more fully. But it is *proof-strategy identical*: ordinary induction on the finite proof sequence. No genuinely different derivation method was demonstrated.

## 6 Why the experiment is not evidence of blind discovery

Five evidentiary boundaries must be kept separate:

1. **Source exposure:** The complete author proof had appeared earlier in the same conversation while selecting and inspecting the theorem. Subsequent requests to "hide the ending" did not erase that prior exposure. The actual test was therefore not blind.
2. **Execution identity:** The role names Player-0, P, R, D, I, C appeared as a requested in-chat representation. There is no evidence here that six independent ATP processes executed under the designated AMLD+0 runtime, or that a distinct authenticated ATP identity was bound to "I" in (1).
3. **Verification:** The final-audit PDF describes conventional mathematical checking of the 45.2 manuscript and explicitly excludes a mechanically accepted proof certificate. The in-chat proof received a mathematical argument review, not an independent external verifier acceptance receipt [2].
4. **Charged costs:** No frozen numerical budget, random seed, exhaustive action trace, or resource ledger for reflection, communications, generation, and checking was produced. Statements about solver speedup, cost reduction, or awareness-assisted performance are unsupported.
5. **Causal attribution:** The generated solution used the same strategy as the original. The episode cannot identify any additional effectiveness caused by AMLD+0 federation, DATA MIND awareness, advanced awareness, or Basin Escape.

The self-referential future claim (1) raises an additional distinction: finding a valid proof of $T$ is mathematical completion, while settling the specific future-tense claim requires evidence that the *designated* prover actually performed the completion under a fixed interpretation of "I", within its authorized run and verifier regime. Merely asserting "I will" is not a certificate.

## 7 What the prior mathematical audit says—and does not say

The October 13 final integrated mathematical audit of Version 45.2 reports *conditional mathematical acceptance for author review*, describing repaired closure and simulation arguments and reporting that 12 of 12 finite toy-model assertions and 9 of 9 structural checks passed. It expressly distinguishes these checks from general theorem certification and notes that no Lean, Coq, Metamath, or independently certified proof-checker artifact accompanies the manuscript [2]. Those findings bear on the source's status, not on independent acceptance of the present in-chat solution.

The additional comparison in this episode is a *mathematical review of a candidate against an existing argument*. It is not a second, independent ATLAS executable audit and should not be represented as one.

## 8 Scientific value and next replication requirements

This result is worth documenting for its *protocol engineering*. It demonstrates how to select an actual source theorem, preserve theorem assumptions, freeze an exact partial-proof handoff, interpret an induction hypothesis in local scope, permit alternative continuations, and explicitly separate a plausible mathematical completion from the stronger claims of blinded discovery and certified machine settlement.

A scientifically controlled successor run would require the following, all fixed *before* the attempt:

1. A new theorem or proof whose withheld suffix is not present in the settler's context, accessible sources, memory, or tools; independently establish that lack of access.
2. The actual frozen original source and exact visible prefix, with complete source-version and handoff hashes; delimit the prefix by reproducible bytes or formally numbered proof steps.
3. An authenticated designated prover identity for "I," with authorized roles and a genuinely separate object-level checker; require accepted proof output, not an unsupported halt or declaration.
4. A declared search budget, costs of all actions (including receipt processing and federation activation), stopping rules, seeds, and complete event trace.
5. A sealed independent evaluation key, opened only after the attempt; accept any mathematically valid proof, not only word-for-word reproduction of the original ending.
6. If awareness is to be tested, matched baseline conditions that differ only in their authorized awareness inputs or controller policy, with predeclared outcomes. A single completion cannot establish a speedup or an awareness effect.

These are *recommendations for a future protocol*, not unannounced modifications of the completed episode or of the previously frozen mathematical target.

## 9 Conclusion and publication-status statement

**Mathematical result:** SOUND CANDIDATE COMPLETION.
**Relationship to source:** same induction argument, different wording.
**Blindness:** NOT ESTABLISHED (source ending previously visible).
**Authenticated AMLD+0 run:** NOT ESTABLISHED.
**External proof-verifier receipt:** NOT AVAILABLE.
**Novel proof method or awareness advantage:** NOT DEMONSTRATED.

The episode succeeded as a concrete exercise in frozen-prefix proof completion and source-to-candidate mathematical comparison. Its importance is also diagnostic: it exposes exactly why a mathematically correct in-chat completion is not automatically evidence of blind proof discovery, active multiagent execution, or verifier-certified self-prediction. Preserving these distinctions makes a later truly blind, charged, independently checked AMLD+0 experiment more interpretable.

# A Author's original continuation, disclosed only for post-test comparison

This section intentionally reproduces the *withheld* portion of the author's original argument. It is provided for public audit of this completed episode and **must not be included in any future settler's blind handoff**.

> If $\pi(j) \in \Gamma$, then $\pi(j) \in W$. Otherwise $\pi(j)$ is a direct consequence of $\{\pi(1), \ldots, \pi(j-1)\}$, and all of those premises lie in $W$ by the induction hypothesis. Since $W$ is inference-closed, $\pi(j) \in W$.
>
> Thus every line of $\pi$ lies in $W$, so in particular $\psi \in W$.

The text is reproduced from the source proof only after the fixed cutoff. Compare this with the candidate case analysis in the main text; the inference is the same.

# B Artifact identity and disclosure policy

Canonical source title: *Depths of a Simulation* 45.2, integrated source, October 13, 2026. Source archive's separate final integrated mathematical audit: four-page PDF, October 13, 2026. Experiment's frozen handoff revision: `Depths45_2_Induction_UNARMED_FROZEN_v2_All_Theorem_Hypotheses.md`. Its source SHA-256 and canonical cutoff are given in Section 3. The paper is a *retrospective account*, not itself a settler prompt, accepted machine certificate, execution log, or published website record.

No claim is made of sentience, phenomenological experience, autonomous preference, or continuity of a subjective identity. Formal self-reference in a settlement conjecture is treated as a typed experimental specification.

# References

[1] B. Tenneson, *Depths of a Simulation: Computers, Formal Systems, Simulations, and Automated Theorem Proving*, Version 45.2, integrated LaTeX manuscript, October 13, 2026, Section 2 ("Induction principle on proofs").

[2] *Depths of a Simulation 45.2: Final Integrated Mathematical Audit*, mathematical review of the repaired original LaTeX manuscript, October 13, 2026, 4 pages. Review assistance: ChatGPT.

[3] Frozen source-handoff manifest, `Depths45_2_Induction_FROZEN_MANIFEST_v2.json`, October 17, 2026; identifies source, cutoff, word fraction, and digest of the submitted handoff. Experiment correspondence and in-chat candidate proof of October 17, 2026.